Xiaoyue (Lucy) Liu

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Education

National University of Singapore, Singapore Aug 2023 – Jan 2027
Bachelor of Engineering (Honours) in Biomedical Engineering | GPA: 4.69/5.0

- **Coursework:** Discrete Structures and Data Analysis, Programming Methodology, Fundamentals of Signal Processing

Vanderbilt University, United States Jan 2026 – May 2026
Undergraduate Exchange Student | GPA: 3.73/4.0

- **Coursework:** AI in Biomedical Engineering, Machine Learning and Natural Language Processing in Healthcare

Awards and Honors

Dean's List, National University of Singapore Jan 2026

- Recognized for outstanding academic performance.

MICCAI Society Registration Grant, MICCAI 2026 Jun 2026

- Selected to receive the MICCAI Society Registration Grant to support attendance at MICCAI 2026 in Strasbourg, France.

Research Experience

Undergraduate Research Intern, Vanderbilt University Jan 2026 – May 2026
Supervisor: ***Professor Bennett Landman****, Department of Electrical and Computer Engineering*
Co-supervisor: ***Professor Louise A. Mawn****, Department of Ophthalmology and Visual Sciences*

- Implemented an anatomical topology-aware loss function for MRI orbital segmentation to support longitudinal volume analysis of thyroid eye disease patients pre- and post-treatment **[2] [3] [8] [10]**.
- Improved anatomical topology-aware segmentation quality, increasing structural accuracy by up to 23.57% over baseline model.

Undergraduate Research Experience, National University of Singapore Oct 2024 – Nov 2025
Supervisor: ***Professor Lei Li****, Department of Biomedical Engineering*

- Introduced differentiable contour rendering for sparse multi-view CMR contour supervision, improving boundary alignment across cardiac views.
- Developed dual-context temporal modeling to preserve cardiac motion and reduce mesh jitter, achieving SOTA cardiac mesh reconstruction **[1] [4] [5] [7] [9]**.

Undergraduate Research Intern, National University of Singapore Jan 2024 – Sep 2024
Supervisor: ***Professor Qianqian Ni****, Department of Diagnostic Radiology*

- Reviewed and synthesized research on AI-driven excipient optimization and pharmaceutical formulation design.
- Gained exposure to experimental techniques and imaging-related workflows in biomedical research.

Publications

Accepted Publications

[1] **Xiaoyue Liu**, Xiaohan Yuan, Mark Y. Y. Chan, Ching-Hui Sia, and Lei Li*. "*CineMesh4D: Personalized 4D Whole Heart Reconstruction from Sparse Cine MRI*." Sep 2026
MICCAI 2026. **[Early Accepted, Top 9%]** **[Paper]**

[2] **Xiaoyue Liu**, Zhenhan Lin, Adam M. Saunders, Gaurav Rudravaram, Isaac Bakis, Melissa Goldin, Justin Shi, Louise A. Mawn*, and Bennett A. Landman*. "*Multi-Object Topology-Aware Anatomical Modeling for Consistent Orbital Segmentation*." Jul 2026
SPIE Medical Imaging 2027. **[Abstract]**

[3] Zhenhan Lin, Adam Saunders, Isaac Bakis, **Xiaoyue Liu**, Bennett Landman*, and Louise A. Mawn*. "*Normative Volumes of Orbital Structures from AI-enabled CT Analysis*." Apr 2026

American Society of Ophthalmic Plastic and Reconstructive Surgery, ASOPRS 2026.

[4] Minghai Shi, Xiaoxian Zhang, **Xiaoyue Liu**, Fan Yang, and Lei Li*. "*DepthScopy: Decoupling Frequency for Endoscopic Depth Estimation in Sparsely-Textured Regions*." Jan 2026

CVPR 2026 Findings Track. **[Paper]**

[5] **Xiaoyue Liu**, Xicheng Sheng, Xiahai Zhuang, Vicente Grau, Mark Y. Y. Chan, Ching-Hui Sia, and Lei Li*. "*Personalized 4D Whole Heart Geometry Reconstruction from Cine MRI for Cardiac Digital Twins*." Sep 2025

Digital Twin for Healthcare, MICCAI workshop 2025. **[Oral]** **[Paper]**

[6] **Xiaoyue Liu**, Liulu Xie, and Wei Chen*. "*Machine Learning-Driven Optimization of Functional Excipients and Their Biointeractions in Drug Formulations*." Sep 2025

ACS Pharmacology & Translational Science. **[Paper]**

[7] Yilin Lyu, Fan Yang, **Xiaoyue Liu**, Zichen Jiang, Joshua Dillon, Debbie Zhao, Martyn Nash, Charlene Mauger, Alistair Young, Ching-Hui Sia, Mark Y. Y. Chan, and Lei Li*. "*Personalized 3D Myocardial Infarct Geometry Reconstruction from Cine MRI with Explicit Cardiac Motion Modeling*." Sep 2025

Digital Twin for Healthcare, MICCAI workshop 2025. **[Oral]** **[Paper]**

[8] Zhenhan Lin, Adam M. Saunders, **Xiaoyue Liu**, Trent Schwartz, Gaurav Rudravaram, Isaac Bakis, Melissa Goldin, Louise A. Mawn, and Bennett A. Landman. "*Corrective Refinement with Cascaded nnU-Net for Through-Plane Resolution Harmonization in Orbital CT Segmentation*." Oct 2026

SPIE Medical Imaging 2027. **[Abstract]**

Under Review

[9] **Xiaoyue Liu**, Dongcheng Cang, Xiaohan Yuan, Mark Y. Y. Chan, Ching-Hui Sia, and Lei Li*. "*Personalized 4D Whole-Heart Mesh Reconstruction from Cine MRI via Multi-Scale Temporal Modeling and Differentiable Contour Rendering*." Jul 2026

Medical Image Analysis. **[Major Revision]** **[arXiv]**

[10] Zhenhan Lin, Adam Saunders, Melissa Goldin, **Xiaoyue Liu**, Justin Shi, Junjie Ma, Shayna Melvani, Myriam Cordey, Ann Xi, Bennett Landman*, and Louise A. Mawn*. "*Orbital Muscle Volume Reduction Correlates with Intraocular Pressure Reduction After Teprotumumab in Thyroid Eye Disease*." Mar 2026

American Academy of Ophthalmology Annual Meeting, AAO 2026.

Internship

MicroPort Medical (Shanghai) Co., Ltd. May 2025 – Aug 2025

Research & Development Intern – Cochlear Implants Department

- Developed cochlear implant noise detection and signal processing pipelines with AI acoustics research team, improving noisy-speech recognition accuracy by 25.3%.

Academic Service

MICCAI 2026, Medical Image Computing and Computer Assisted Intervention 2026

- Served as a reviewer for MICCAI 2026.

Skills

Programming: Python, Java, MATLAB

Machine Learning Frameworks: PyTorch, TensorFlow